

SAJIL S NAIR

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Summary

Computer Science and Engineering (CSE) student with hands-on experience building secure API systems and AI-powered web applications. Proficient in Python, Java, C++, C, SQL, and JavaScript, with project experience in API gateways, backend development, machine learning, and web technologies. Experienced with FastAPI, Kong, Keycloak, Docker, React.js, Node.js, embedded systems, and AWS fundamentals. Seeking an entry-level software engineering role or internship.

Technical Skills

Programming Languages: Python, Java, C, C++, SQL, JavaScript

Web Technologies: HTML, CSS, React.js, Node.js

Backend & Security: FastAPI, REST APIs, Kong, Keycloak, JWT

AI/ML & 3D: Machine Learning, Sentiment Analysis, Jupyter Notebook, Blender, glTF/GLB, A-Frame, WebXR

Tools & Platforms: Docker, Docker Compose, Git, GitHub, VS Code, AWS Fundamentals, Arduino, RFID

Projects

NMS-API-Access-Control

Kong, Keycloak, FastAPI, Docker, JWT, REST APIs

- Built a containerized API gateway layer for NMS Northbound APIs using Kong and a FastAPI mock NMS, centralizing request routing and access control.
- Integrated Keycloak-based JWT authentication and protected NMS routes, with Kong enforcing token validation and rate limiting.
- Orchestrated Kong, Keycloak, and the mock NMS as Docker Compose services on a shared network, keeping the NMS backend internal.

MindPulse AI

React.js, Node.js, NLP, Sentiment Analysis

- Built a web-based weekly employee feedback system covering stress, workload, emotional state, and written comments.
- Applied NLP-based sentiment analysis and structured scoring to classify burnout risk as Low, Moderate, or High.
- Implemented employee and manager dashboards with personalized reports, aggregated team analytics, and historical trend tracking.

AR Interactive 3D Heart Model

Blender, GLB, model-viewer, WebXR

- Created an animated 3D heart with dual display panels in Blender and exported an optimized GLB asset for web and AR.
- Built a browser viewer with model-viewer, camera controls, autoplay, and AR launch via WebXR, Scene Viewer, and Quick Look.

VR Interactive Room

A-Frame, JavaScript, GLB

- Developed a browser-based virtual room with A-Frame and a GLB environment for first-person exploration.
- Implemented keyboard/touch movement and rotation with raycasting-based collision checks against room geometry.

RFID Attendance System

Arduino, RFID, Arduino IDE

- Built an Arduino-based RFID attendance system to identify users and record attendance in real time.
- Automated attendance capture to reduce manual entry and improve record consistency.

Education

B.Tech in Computer Science and Engineering (CSE)

Expected 2027

Reva University

Certifications

- Machine Learning using Python — Simplilearn, 2025
- Artificial Intelligence — 1Stop.AI
- Getting Started with AWS Services Fundamentals for Beginners — Simplilearn, 2026
- Introduction to Computer Networking — Simplilearn, 2026
- ReactJS for Beginners — Simplilearn, 2026
- Getting Started with MS Excel — Coursera, 2023